



LESSON - 16

SIKHOGE KYA???

INTRODUCTION & KEY POINTS

- INTRODUCTION
- OVERVIEW OF DATABASES
- APPLICATIONS & JOBS
- SOFTWARE INSTALLATION -
- MYSQL SERVER AND MYSQL WORK BENCH WITH DATABASE SETUP
- TABLES CREATION & FIRST QUERY

SQL STATEMENT FUNDAMENTALS

- SELECT * FROM
- SELECT DISTINCT
- COUNT, ALIAS AS
- SELECT WHERE
- LIMIT AND ORDER BY
- OPERATORS AND FILTERING
- BETWEEN, IN, LIKE AND ILIKE
- CASE IF ELSE
- COALESCE, NULLIF



DATA MANIPULATION (DML)

- INSERT
- UPDATE
- DELETE
- COPY RENAME STRUCTURE
- CHECK CONSTRAINT

DATA DEFINITION LANGUAGE (DDL)

- DATA TYPES & ROWS COLUMNS
- PRIMARY FOREIGN & OTHER KEYS
- CREATE TABLE
- ALTER TABLE
- DROP TABLE
- CONSTRAINTS

JOINS & SUB QUERY

- INNER JOIN
- LEFT JOIN (LEFT OUTER JOIN)
- RIGHT JOIN (RIGHT OUTER JOIN)
- FULL JOIN (FULL OUTER JOIN)
- CROSS JOIN
- SELF JOIN
- -----
- UNION, UNION ALL, MINUS, INTERSECT
- STRING FUNCTIONS AND OPERATORS
- SUBQUERY - IMPORTANT
- DATE & TIME FUNCTIONS

GROUP BY & HAVING

- INTRODUCTION TO GROUP BY
- AGGREGATION FUNCTIONS
- GROUP BY - CHALLENGE
- HAVING
- HAVING - CHALLENGE TASKS

ADVANCE SQL

- VIEWS / TEMP TABLES & INDEXES
- FUNCTIONS & TRIGGERS
- STORE PROCEDURES
- TRANSACTIONS
- WINDOW FUNCTIONS
- QUERY ANALYSIS TECHNIQUES

Group By - Imp!

The GROUP BY clause is used to arrange identical data into groups. It is often used with aggregate functions like COUNT(), SUM(), AVG(), MAX(), and MIN() to perform operations on each group of data.

Syntax:

```
SELECT column1, column2, AGGREGATE_FUNCTION(column3)
    FROM table_name
        WHERE condition
            GROUP BY column1, column2
                ORDER BY column1;
```

Having - Imp!

The HAVING clause in SQL is used to filter query results based on aggregate functions. Unlike the WHERE clause, which filters individual rows before grouping, the HAVING clause filters groups of data after aggregation. It is commonly used with functions like SUM(), AVG(), COUNT(), MAX(), and MIN().

Syntax:

```
SELECT column1, aggregate_function(column2)
    FROM table_name
        GROUP BY column1
            HAVING condition;
```

Difference between WHERE and HAVING

- WHERE is used to filter rows before grouping.
- HAVING is used to filter groups after aggregation.